



The ethics of erroneous AI-generated scientific figures

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Abstract

The number of AI-generated figures in scientific publications is increasing, unfortunately leading to high-profile retractions of papers featuring inaccurate visualizations. The lack of definitive guidelines for AI-generated scientific and educational visualizations results in several ethical issues and dilemmas. At the same time, we maintain that there should not be a double standard regarding the factual correctness of figures only due to AI involvement in their creation and argue in favor of measured responses. We present a framework considering the communicative purpose of a visualization, the type and function of the figure in a paper, the type of error, risks, and the appropriateness of the figure as a means to support decisions regarding the severity of issues of AI-generated images for scientific and educational aims. By outlining a more fine-grained analysis of error types and visualization characteristics, we provide orientation for the current controversy surrounding AI-generated figures. This framework can also serve as a starting point for considerations regarding AI use by students. In addition, we discuss more sophisticated ways of using AI systems to generate visualizations that avoid the pitfalls of general-purpose text-to-image tools.

Keywords Illustration · Ethics · Artificial intelligence · Publishing ethics

Introduction

In 2024, two peer-reviewed articles containing AI-generated figures featuring major anatomical inaccuracies (Guo et al., 2024; Wu et al., 2024) were retracted after becoming social media sensations. One figure in the first paper depicts a rat with highly unusual genitalia alongside illegible text (Guo et al., 2024), while the second article includes anatomical visualizations of the human leg, incorrectly showing two bones for the upper leg and three lower leg bones (Wu et al., 2024) (see Engel-Hermann & Skulmowski, 2024, for a discussion of these incidents). Again, the text presented in the figures in the second paper is illegible. Strikingly, the visual quality of the figures in both papers is very high, utilizing the style of traditional, lavishly illustrated anatomy books (Guo et al., 2024) and modern image processing effects (Wu et al., 2024). The guidelines of major publishers regarding

AI use are inconsistent, but many emphasize the authors' responsibility for checking the factual correctness of the AI output they wish to incorporate in their manuscripts (Gulumbe et al., 2024). After all, the factual correctness of scientific works still is the cornerstone of the scientific enterprise.

However, as we assume that many more researchers will succumb to the temptation of submitting visually appealing, yet easily produced AI-generated figures as part of their manuscripts, it is very likely that similar cases will follow. The events outlined above highlight that, in contrast to texts, there is a lack of clear rules and guidance concerning the factual correctness of figures, leading to various ethical dilemmas. Thus, we wish to discuss the new phenomenon of AI-related inaccuracies and misrepresentations in scientific figures that is in urgent need for more definitive guidelines. In addition, we argue for the adoption of reasonable and fair procedures when dealing with research containing AI-related inaccuracies in illustrative content. These developments also have implications for education, as students should be trained regarding how to use AI tools in appropriate ways. At the same time, there exist no clear criteria for AI-generated visualizations submitted by students.

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Until the aforementioned two high-profile retractions, the debate around misleading images has almost exclusively been centered around cases of image manipulation (e.g., Frow, 2012; also in the context of AI tools, Gu et al., 2022; see also Kim, Srivatsa, et al., 2024), with little discussion regarding errors in illustrative figures. In contrast to cases in which images constitute the primary research data, resulting in a clear reason for retracting papers containing manipulated images, both recent cases mentioned above raise novel questions. The figures in question were supplied as illustrations, presumably with the intention to summarize the points made in the respective article texts.

While extreme cases may be argued to contain so much inaccurate information that they would negatively affect the integrity of the scientific record, less grave cases may require a different response. However, it should be noted that, unlike for scientific writing, no clear guidelines exist regarding thresholds for correctness and accuracy in the creation of scientific illustrative figures, making it difficult to assess AI-generated visualizations in a fair manner. With the advent of inaccurate AI-generated visualizations, researchers have only recently started to develop criteria for the assessment of such visualizations (Currie et al., 2025; Temsah et al., 2024).

While we do not wish to trivialize the severity of using incorrect AI-generated visualizations, it should be avoided to introduce a double standard for the quality of such figures compared to figures produced using “traditional” methods. It is common for papers to feature loose sketches, anatomical visualizations that omit small or inconsequential details, and pictures edited to blur parts containing extraneous information. Since simplifications, modifications for visual emphasis, and minor inaccuracies will likely be found in AI-generated figures in the future, what should be the consequences? Should all AI-generated illustrative figures be held to a higher standard than sketches and diagrams created without AI? Even with future advancements in AI-based image generators, there will be a risk of subtle errors that could remain unnoticed during the peer-review process. This emerging gray area highlights that more definitive guidelines need to be developed.

Unfair punishment?

Before presenting a framework for the assessment of incorrect AI-generated figures, we first want to discuss why people might be more adamant to “punish” authors using AI tools compared to authors who created inaccurate visualizations using other methods (as evidenced by some of the social media debate surrounding the two recent cases; for an overview of the social media response for Guo et al., 2024, see <https://www.altmetric.com/details/159497259>).

As there are several papers containing non-AI-generated inaccurate figures that may merely be perceived as careless or just of sufficient quality to communicate only specific types of information, the context of a figure appears to be a crucial factor in the assessment of AI-generated visualizations (Currie et al., 2025). Their high quality of design, the plausible rendering of shapes and shadows, and the appealing style may result in an increased expectation for factual correctness. A style associated with a high production quality may suggest that a comparable amount of effort has been invested in checking the correctness of the presented information. Klug and Pietsch refer to the style of erroneous AI-generated figures as a “falsely scientific aesthetic” (Klug & Pietsch, 2024, p. 1).

More fundamentally, negative attitudes towards authors that get “caught” using inaccurate AI-generated images may be explainable due to the similarity to plagiarism: while plagiarism is defined as passing off work done by others as one’s own, using a generative AI to create professional-grade illustrations and then failing to check their factual correctness may similarly be perceived as an unethical “shortcut.” Thus, it could be the case that incorrect AI-generated figures come across as more off-putting than inaccurate human-generated graphics, possibly leading to harsher judgments (for overviews of attitudes towards AI-generated content, see Hadan et al., 2024; Skulmowski, 2024).

A framework for assessing AI-generated visualizations

The widespread availability of generative AI tools raises several new ethical questions for anyone involved in the scientific enterprise. While the factual correctness of the scientific record must be preserved, generative AI tools can provide equity at the cost of accuracy. Researchers working in countries without sufficient budgets and resources to allow them to include high-quality illustrations in their work finally have the opportunity to utilize generative AIs as an equalizer (for a related discussion of the productivity-enhancing effects of large-language models, see Noy & Zhang, 2023), with a great potential for increasing the global fairness in publishing. Accordingly, we argue that reasonable policies should be adopted that take various factors into account in dealing with inaccuracies found in scientific publications.

A recently presented framework in the medical context (Temsah et al., 2024) contains the factors *image accuracy*, *image-text usefulness*, *attractiveness to medical professionals*, and *suitability for medical education*. A different paper contains analyses of AI-generated visualizations that are discussed concerning their appropriateness for medical

education (Currie et al., 2025). In these analyses, it was found that currently popular generative AI tools can produce highly detailed, sometimes quite artistic, but ultimately inaccurate images for different exemplary use cases (e.g., heart anatomy, radiographic images, and scan images). Currie et al. applied a broad framework of AI ethics (Floridi et al., 2018) to the field of generative AI in medicine, thereby including ethical principles regarding regulation, copyright, privacy, and liability. Based on these considerations, Currie et al. (2025) arrived at a detailed flow chart in which decisions concerning generative AI systems and their use in the context of medicine are guided by ethical principles. Their chart is based on a collection of criteria including the following: user characteristics (such as whether the user is a medical educator), the aims of the user and their audience, the publication venue, the accuracy of images, and risks for injury and liability.

In order to assess whether an incorrect AI-generated figure poses a serious problem, we have collated a number of

factors that are relevant for scientific publishing and educational purposes with a stronger focus on visualization characteristics than previous frameworks (Table 1): the communicative purpose of the visual element in question, the type of figure, the severity of the deviation from the original, consequences and risks, and the appropriateness of the figure. We will describe these factors in detail in the following sections.

The purpose of the visual element

Firstly, we argue that it is important to distinguish between the different kinds of communicative purposes that figures can fulfill (see Currie et al., 2025; Peterson et al., 2021). Accordingly, errors in figures should be considered with the purpose of the visual illustration in mind. Figures can feature decorative elements that have no direct relevance for the correctness of the scientific content. For example, if an educational visualization of dental anatomy jokingly included a small drawing of the fictional Tooth Fairy next to anatomically correct visualizations, this would probably not result in a violation of scientific principles. Likewise, this image would certainly not mislead adult learners into believing that the fictional character is real. However, the inclusion of purely nonsensical extraneous elements would certainly raise questions. Clearly, such decorative elements should be used sparingly regardless of their method of creation.

Beyond decorative elements, some visual elements in figures have an illustrative purpose. Such illustrations are generally not expected to conform to the highest standards of accuracy, since they are often only included to provide orientation or to act as a visual guide. One example demonstrating that unexpected elements in illustrative figures have been considered as minor issues that merely need to be amended rather than being retracted is an article (Chiou & Bergey, 2018) containing a figure that caught attention (Goodyear, 2018). The article deals with DNA sampling from feces, and the first figure of that paper presents a DNA sequencing method. As an illustrative embellishment, the figure contains a drawing of a baboon sitting next to fecal matter. In the original version of the paper, the excrement featured a drawing that some people perceived to resemble a politician (Goodyear, 2018). In this case, the journal did not issue a correction, but simply updated that part of the article by removing the drawing from the figure (for an overview of the social media response to this incident, see <https://nature.altmetric.com/details/32472137>). The very small size of the “unusual aspects” (as the editor’s note on the article refers to the issue), the simplified style of the figure as well as the fact that the drawing of the excretion does not constitute the main part of the figure could be considered as

Table 1 An overview of the continua using which erroneous scientific figures can be described in order to assess the gravity of one or more inaccurate visual elements, with more benign issues towards the left side and indicators of serious problems towards the right side

Communicative purpose of the figure			
Decorative, with no direct relevance for the content	Illustrative, not aiming at an authoritative depiction	Representational, intending to faithfully depict	Reference, requiring the highest standard for correctness
Type and function of the figure			
Supplemental or extraneous content (e.g., graphical abstract)	Introductory illustration	Method illustration	Main result visualization
Type of deviation from original			
Visual emphasis or stylistic variation	Inaccuracy regarding proportions, shapes, colors, typos in text or labels	Misrepresentation of proportions, shapes, colors, incorrect text or labels	Incorrect additions or omissions of important elements, nonsensical text or labels
Consequences and risks			
None	Confuse readers	Mislead readers, resulting in misconceptions that need to be corrected	Create health risks or legal ramifications
Appropriateness and professionalism			
No professional standards are being violated, no inappropriate content	Minor infractions against professional standards, content that may be inappropriate for some viewers	Major violations of professional standards, content that most viewers find inappropriate	Highly inappropriate, violating professional standards or guidelines
Less severe		More severe	

arguments for the conclusion that the element in question was not of a particularly high importance. It should be noted that the style of the drawing of the baboon clearly indicates that it is meant to serve an illustrative purpose rather than acting as an authoritative depiction. In many cases, figures contain representational imagery that is included with the intention of faithfully representing a real object. This could be the case when presenting an anatomical structure, a medical procedure, a chemical, or any other type of object for the sake of discussion, commentary, explanation, or similar purposes. Such figures often contain photographs, three-dimensional renderings of scan data, drawings, or other image types offering a high level of accuracy and detail. This type of visualization appears in many research papers and is thus likely to raise many questions due to AI-induced inaccuracies in the future.

In our framework, the highest level of scientific accuracy is required for reference works, many of them being educational materials. It should be uncontroversial to expect that textbooks, papers, and other resources provided for educational purposes do not contain errors, regardless of which technology was used to create the figures for such materials. Considering the purpose of a figure, it becomes clear that inaccuracies in decorative visual elements and illustrative figures usually do not pose a threat to scientific integrity and have in previous cases been corrected in case irregularities come to light after publication. However, as readers typically expect the content of representational visualizations to be as thoroughly checked as the text of a publication, errors in such visualizations are likely to trigger more serious responses. Yet, we argue that additional factors should be considered when assessing the severity of an erroneous figure and discuss the different functions that figures can have in a paper in the following section.

Type and function of the figure

Another criterion that can be used to determine whether inaccuracies in an AI-generated figure pose a serious issue warranting consequences is the type and function of the figure in question. Scientific papers can contain various types of figures differing in their importance. Arguably the most important figures in a paper are visualizations of the main results. Errors in such figures are usually the most consequential mistakes, and it may generally be advisable to avoid creating or editing crucial figures requiring utmost correctness with AI tools prone to imprecision. Other kinds of figures include illustrations of the utilized methods. These visualizations need to be accurate enough to allow readers to understand and reproduce the findings, but in many cases, minor simplifications or deviations are unlikely to affect these purposes.

Introductory sections also often contain figures illustrating the subject matter of a paper. It may be argued that errors in such visualizations may not warrant retractions or similarly harsh consequences as long as these figures are not explicitly provided for reference or educational purposes. In addition, there are figure types that are extraneous to the paper, such as graphical abstracts. In particular if such elements are not a part of the paper and only displayed on issue overviews of journals, errors should not be of critical importance and could be fixed rather easily.

In addition, the publication type should similarly be considered in the assessment of AI-generated erroneous visualizations. For instance, clinical guidelines, educational textbooks, and peer-reviewed research papers will need to meet higher standards than other types of publications.

Type of deviation from the original

There are numerous ways in which a visualization can differ from the original. By original, we predominantly mean the real-world object that a visualization is supposed to depict. In some special cases, it is possible that this term could mean other types of reference material, such as visualizations generated using imaging procedures. Depending on how strongly a visualization deviates from the original, such changes can turn out to be problematic. Generally benign visual alterations in a scientific figure consist of small changes to shapes, colors, or proportions; sometimes being the result of a particular visual style and other times being utilized for emphasis. In case that such changes are not applied in an attempt at image manipulation, these changes are quite common, in particular in schematic or simplified drawings (for overviews on different levels of detail, see Quillin & Thomas, 2015; Skulmowski et al., 2022). Thus, such modifications should be equally accepted if an AI has been used to generate an image.

Inaccuracies concerning proportions, shapes, and colors, however, raise more serious questions, in particular for representational and reference work visualizations. Especially for AI-generated visualizations produced in a highly detailed and realistic style that may lead readers to accept the visual information at face value (Engel-Hermann & Skulmowski, 2024), inaccuracies may be hard to defend. Ultimately, the context will be relevant to decide whether a minor inaccuracy constitutes a problem. By minor inaccuracy, we mean small deviations that alter the shape, proportion, or color in a way that deviates from accepted knowledge, but is inconsequential for most purposes. For instance, it may be necessary for an anatomical visualization to accurately capture whether the inner or outer condyle of the knee is larger, but outside of the medical context, a failure to consider this

comparatively minor aspect of proportion in a visualization of the femur is likely to be irrelevant.

However, as soon as such inaccuracies misrepresent the subject matter in a more fundamental way, it is likely that a correction or other measures will be necessary to rectify the scientific record. Thus, a misrepresentation in our system is not merely a small deviation, but a large or at least highly meaningful aberration. As we have seen in the case of the paper containing a figure depicting a rat with enormous genitalia (among other errors) (Guo et al., 2024), a retraction was deemed appropriate by the journal. Furthermore, additions or omissions of key visual elements constitute major errors and can easily happen when researchers use AI tools to generate visualizations (Currie et al., 2025). Still, it should be noted that the severity of omissions is highly context-dependent. In an educational diagram of the heart, it may be critical to include even the smallest veins if this image is supposed to serve as a comprehensive reference, while this level of precision may be too overwhelming for a diagram in an introductory lesson on heart anatomy. However, if an AI was to invent veins that do not exist in reality, this is likely to be a major issue. Thus, we argue that additions can be considered highly problematic in most cases.

The exhaustive description of an ethical issue in publishing requires a consideration of the role of the author(s). First and foremost, it is the author's responsibility to ensure that the content they supply meets the high standards of scientific publishing. Still, it should be noted that different kinds of errors in scientific figures can be the result of a confusion regarding different sets of standards to which authors may or may not adhere. One of the more innocuous ways an author could violate their responsibility to ensure the correctness would be a failure to notice minor deviations in AI-generated visualizations, such as slight simplifications or minor errors in proportion. One could argue that overlooking minor deviations that do not confuse or mislead readers is not something that authors should be held accountable for. In line with this reasoning, less severe measures would be an appropriate course of action, such as updating or correcting an inaccurate figure.

However, tolerating rather than failing to notice deviations could be argued to be a quite different situation in which an author willingly accepts the risk to confuse or mislead readers. Such behavior is in conflict with the established ethical principle of non-maleficence that has been identified as a central component of AI ethics (Floridi et al., 2018). Still, as the consequences of minor inaccuracies will likely be less severe, the action to be taken in such a case should still be proportional and, as we argue, should not be used to "punish" authors. Since it may often be difficult or impossible to determine whether an author failed to notice an inaccuracy or whether they knowingly tolerated

it, we argue to be cautious in judging such minor errors. In contrast, tolerating major misrepresentations (or failing to check for errors of this magnitude) certainly endangers the academic integrity and scientific correctness of a publication, which likely will require a correction or retraction.

In summary, inaccuracies in AI-generated scientific visualizations may be present in embellishments without a direct connection to the subject matter of a publication, appear as minor or major inaccuracies, or as the addition or omission of relevant visual elements. It will need to be determined at which point of the spectrum between minor and major inaccuracies researchers can correct a visualization instead of retracting the entire publication. Again, this decision will likely be affected by additional context factors, such as the risks that a specific inaccurate visualization poses as described in the following section.

Consequences and risks

While we do not propose a purely consequentialist assessment, we deem the risks that could arise from inaccurate AI-generated scientific figures as a relevant factor in the assessment of such visualizations (in line with Currie et al., 2025). For many kinds of minor inaccuracies, in particular if they appear in visual elements unrelated to the main topic of a visualization, there will very likely be no risk to readers. It could be argued that such minor deviations do not result in any negative consequences and thus do not require any action.

In case that the worst conceivable consequence would be that a figure may confuse readers regarding information that is not in conflict with the main text of a publication, we suggest that a correction should be issued in the case of scientific publications. We expect that the majority of future cases of inaccurate AI-generated illustrative figures will fall into this category. Especially in cases in which the main text is scientifically sound, it may be sufficient to remove incorrect illustrative figures or to give the authors an opportunity to correct them. In contrast to cases of image manipulation, this action could be considered to be sufficient to correct the scientific record.

A more severe consequence would arise if a visualization was able to profoundly mislead readers, resulting in misconceptions that need to be corrected. In such a case, additional consequences such as health risks (e.g., injuries) or legal ramifications may arise (Currie et al., 2025). Depending on the effort required to correct the resulting misconceptions, a correction or retraction of a scientific publication may be warranted. In the case of educational resources, misleading materials should obviously not be used, regardless of whether an AI was used to create that material.

We suggest that retractions should be limited to cases in which a high risk for misleading readers concerning key aspects of the paper is to be assumed and article removals may be appropriate in cases in which illustrations misrepresent physical procedures in a manner that poses a risk when acted upon (which already is the standard procedure for potentially dangerous non-AI content). However, generally we would find it difficult to accept that the text presented in AI-generated figures is allowed to contain erroneous information, as texts and labels can easily be edited even by novices in graphic design and thus should meet the accepted standards for the publication of scientific writing.

Appropriateness and professionalism

Lastly, we wish to highlight that AI-generated visualizations can be inappropriate and unprofessional (similar to the suitability criterion proposed by Temsah et al., 2024). This can be the result of a variety of issues. A striking example would be violations of professional codes of ethics. For instance, AI tools may be trained on source materials that were obtained in an unethical or illegal manner, including highly personal images (see Cornwall et al., 2024; Katirai et al., 2025). The use of such tools in itself and by extension publishing outputs generated in this manner could violate professional codes of ethics, legal rights, as well as legal agreements with publishers (Cornwall et al., 2025). Similarly, other types of AI-generated images can be highly inappropriate, for instance if they contain indecent imagery. As a result, there are field-dependent requirements for visualizations that AI-generated visualizations need to fulfill in order to become acceptable.

Solutions

While we do not wish to present AI-related inaccuracies as less problematic than other types of scientific errors, it may be naive to hope that all researchers and educators will resist the use of AI tools to generate visualizations. Given the current state of this technology, this trend will likely introduce at least some level of inaccuracy (Buzzaccarini et al., 2024; Currie et al., 2025; Noel, 2024; Temsah et al., 2024), and several measures can be taken to mitigate the negative consequences of this transformation.

In general, a notice that an AI was used to create a figure could provide the necessary context to interpret the factual correctness with some level of caution. There is an ongoing debate around adjusting the expected level of academic honesty to the new reality of AI-assisted content generation (Wang & Yokoyama, 2025; Eaton, 2023) and while this debate has largely been focused on generating text, more

specific guidelines for the use of AI in the creation of figures is needed (Kim, Um, et al., 2024). In particular, we need to be prepared for AI-enhanced functionalities being embedded into standard image editors, making it difficult for users to keep track of using AI features. A particularly difficult problem arises through software that lets users edit existing images using AI functionalities (commonly referred to as “inpainting,” see Klug & Pietsch, 2024). Such modifications blur the boundaries between non-AI and AI-generated content. In line with recommendations to disclose AI use in writing and editing manuscripts in instances in which AI tools meaningfully alter the text (Hosseini et al., 2023), the utilization of such functionalities in the manipulation of visualizations should be acknowledged as well (see Kim, Um, et al., 2024; Klug & Pietsch, 2024). The development of clear guidelines on how to disclose AI use in the creation of visualizations is an urgent gap that needs to be closed.

Suggestions for AI-based misconduct detection are currently being worked on (Hosseini & Resnik, 2025) and should include a fair and balanced guide for assessing inaccurate AI-generated images. Using systems such as our list of continua (see Table 1), it would be possible to assign numerical values to the different levels of severity in which a visualization can violate ethical standards. Based on such a system, a score could be computed to guide journals and editors on how to react to issues found in AI-generated visualizations post-publication, thereby generating a more objective assessment that may help to avoid unfair punishment as well as too lenient interpretations. However, different research fields and their journals will need to determine appropriate values and thresholds for consequences that may need to differ between fields.

Another solution would be the recommendation to use a style for AI figures that does not imply an authoritative, definitive representation, but rather a more “sketchy” or loose style that communicates a level of uncertainty. This measure may soften some of the high expectations that the professional style of AI-generated visualizations may evoke (see Slater et al., 2020, for a related suggestion in the context of virtual reality).

The recent developments introduce an additional conundrum, namely the responsibilities of reviewers, editors, and perhaps even supporting staff to detect mistakes. Again, we do not wish to load off the responsibility for the factual correctness of figures to reviewers, editors, or even typesetters, but with the new challenges introduced by technological advances, it may be necessary to add specific AI-related quality checks to the entire publication process (see Kim, Um, et al., 2024). One issue needed to consider when implementing such checks is the variable detectability of different types of mistakes. While an editorial assistant could easily spot “gibberish” text in AI-generated figures,

other inaccuracies will require expert knowledge. While the authors of a paper are ultimately responsible for the content they submit for publication, reviewers and editors alike should be aware of their responsibility to keep the literature free from inaccurate, let alone nonsensical AI-generated visualizations (Klug & Pietsch, 2024). Thus, it would be advisable to let all people involved in the publication process contribute to AI-related quality checks to minimize the likelihood of repeating the kinds of cases described above.

It has previously been suggested that erroneous AI-generated visualizations could be even more of an issue in the context of some business models found in the world of academic publishing, most notably fee-based open-access publishing promising rapid publication (Klug & Pietsch, 2024). Related to this point, it will generally be undesirable for publishers to admit that their editorial procedures have failed to notice inaccuracies or severe errors brought about through AI use. This conflict of publishers and editors could result in various negative consequences. While a strict enforcement of AI-related guidelines, resulting in corrections and retractions, could restore and maintain the integrity of the scientific record, thereby potentially building trust in the scientific enterprise, large increases in retractions could backfire. The public might gain the impression that scientists are wasting taxpayer-funded resources while producing subpar research. Conversely, ignoring grave mistakes and thus tarnishing the scientific record is likely to erode trust in the scientific enterprise as well. As a consequence, publishers and editors will need to come up with transparent guidelines that are able to address unethical practices and incorrect information while avoiding to exaggerate bagatelle cases.

In addition, some approaches to publishing may be subject to fewer legal ramifications than commercial publishing operations. As some generative AI tools restrict the use of their outputs to noncommercial purposes, some of the recently proposed alternatives to commercial publishing (e.g., Brembs, Huneman, et al., 2023; Brembs, Lenardic, et al., 2023) may have a higher compatibility with such restricted outputs. However, even under such circumstances, other issues such as the ethical problems of attribution and plagiarism remain (Klug & Pietsch, 2024; for an overview, see Cornwall et al., 2024).

Outlook

The problems discussed in this article pertain to the utilization of novel text-to-image generators that can create professional-looking, but inaccurate visualizations using commands entered in the form of ordinary language and without the need for specialized computer skills. As a

Table 2 Overview of an AI-based image generation pipeline allowing checks for correctness

Steps involved in AI-based image generation involving checks

1. Obtain correct source material in the form of reference images or three-dimensional models, check for correctness
2. Use photogrammetry or other image-to-3D tool to generate a model, validate the correctness of the model
3. Use manual effort (e.g., retopology) or AI-based functionalities to optimize the geometry or to remove artifacts and errors (if necessary)
4. Create renderings
5. Use the render output as the basis for further AI-based optimization of the aesthetics and visual design

result of the media hype regarding this particular technology, other tools allowing the (semi)automated production of visualizations that may be more appropriate for the production of visualizations have not been getting the attention they deserve. This has resulted in the current state in which researchers and educators often choose to sacrifice control for ease-of-use.

Fortunately, besides effortful methods such as photogrammetry that require dozens of photographs of an object in order to generate a three-dimensional (3D) model (Petriceks et al., 2018), other alternatives to text-to-image AI tools have emerged. There are several related AI systems that allow the creation of 3D models from just one or a few photographs (Ye et al., 2024). Other AI tools enable the optimization of the visual design of images from an aesthetic standpoint and other more specialized purposes. Instead of utilizing general-purpose text-to-image AI tools that in many cases may not have been trained on a sufficiently large corpus of (vetted) subject-specific images, we emphasize the need for more sophisticated content creation pipelines. Researchers and educators should not blindly follow the AI hype and use widely reported tools, but rather develop and validate new methods for visualization that give them opportunities for correcting inaccuracies and errors (Klug & Pietsch, 2024; Engel-Hermann & Skulmowski, 2024).

These new pipelines will require slightly more knowledge regarding the tools utilized in the visualization process, but can contribute towards the correctness of the final output with just slightly more time and effort invested. However, instead of relying on AI tools using short prompts without the possibility to make precise corrections, authors and educators should utilize AI systems in a manner that keeps them in control. In case they have basic knowledge of 3D tools, an exemplary pipeline for tool-assisted content creation in science and education could look like the suggestion presented in Table 2.

In this proposed pipeline system for the generation of scientific and educational visualizations, the first step would be to collect correct and appropriate reference material or source files. This could be achieved by compiling reference

images with permissive licenses and clear rules for attribution, by taking photos, or by using scan data. The reference material should be checked for correctness and appropriateness, and trusted databases should be used. From this material, photogrammetry or AI-based image-to-3D tools can be used to create models using only a single picture. Of course, this step also requires careful checks regarding inaccuracies introduced by the process. Using 3D software such as Blender (<https://www.blender.org>), corrections to the resulting mesh can be performed given some familiarity with the software. This step will likely be simplified or automated by AI tools in the future as well. In any event, creating a two-dimensional image using a virtual camera and lights for the rendering process can rather easily be learned. The resulting renderings could be submitted to additional AI processing, for instance in order to enhance the aesthetics and visual design of the images. Again, careful checks regarding inaccuracies are necessary. In the future, the effortful intermediate step of creating a 3D mesh, geometry clean-up, and rendering could possibly be avoided through the use of improved AI tools that allow more control over the generated images. Until then, more effort will need to be invested into the AI-assisted image generation process to ensure correctness. Currently, a more straightforward method would be an iterative process using appropriate base images that are modified using AI-assisted inpainting as proposed by Klug and Pietsch (2024). In sum, we wish to highlight that there are numerous tools to assist in the creation of visualizations that do not take away control over finer details.

Although not constituting the main point of our argument, we nevertheless wish to briefly discuss the ethical and legal prerequisites for using AI-generated images. Generative AI tools open up a difficult legal and ethical gray area in which it may not always be clear whether one is allowed to use AI output due to copyright infringement (Cornwall et al., 2025; Samuelson, 2023) and the problem of attribution (Khosrowi et al., 2024). For many subject areas, there are even more ethical issues to be considered, such as the provenience of the source materials (Cornwall et al., 2016; Cornwall et al., 2025) and the required consent (Cornwall et al., 2024; Deane & Byers, 2024). Ideally, generative AI tools should only be trained on data made available with permissive licenses (Samuelson, 2023) that was ethically sourced (Cornwall et al., 2024). Even if these conditions are met, there still will be issues to be resolved within academic publishing, such as the proper attribution of AI-generated visualizations and training material (Khosrowi et al., 2024).

At all levels of education, the introduction of generative AI tools will amplify the need for concrete principles for assessing student-generated visuals. Many papers discussing educational approaches revolving around drawing remain rather vague concerning how to assess student work,

often merely suggesting to rely on rubrics (e.g., Quillin & Thomas, 2015). We highlight that expectations concerning visualizations in science are interconnected with educational practice: it will be difficult to demand the utmost degree of accuracy from scientific figures if schools and universities lower their expectations concerning correctness as a result of students' adoption of AI tools. Likewise, students at all levels of education would be negatively affected by lax standards, leading them to be exposed to inaccurate AI-generated contents.

Conclusion

In summary, we anticipate a wave of publications featuring inaccurate AI-generated figures and argue that there should be a measured response taking into account the communicative purpose of the image, the type of the figure, the type of error(s), consequences arising from the error(s), and the appropriateness of the figure. Given the great potential that AI holds for scientific visualization, there should not be a knee-jerk reaction to retract such papers simply because the authors used an AI. Specifically, we do not find it reasonable to "punish" researchers for minor inaccuracies in AI-generated illustrative visualizations, in particular if they disclose the use of an AI and take additional steps that avoid the perception of the images as authoritative. Combined with an increased public awareness that AI-generated visualizations may contain inaccuracies, we consider it reasonable to tolerate minor inaccuracies that would have been acceptable in sketches and simplified diagrams created without an AI. Otherwise, there would be a risk for the introduction of a double standard.

However, we see the need for establishing a more precise threshold for the accuracy expected from (illustrative) scientific visualizations, regardless of their creation method. While academic training involves instruction regarding how to write correct and original texts, a similar emphasis concerning the production of figures is usually not found. Our proposed system of error continua could serve as the starting point for using AI in several educational fields and at many levels. This system of the severity of inaccurate or otherwise problematic AI-generated visualizations can help to arrive at more objective and comparable measures, potentially through the calculation of integrity scores. In addition, we recommend to use corrections instead of retractions for most cases in which minor or inconsequential errors come to light after publication and suggest to include AI-related quality checks at all stages of the publication process. We also assume that given the increased normalization of AI use, erroneous AI-generated visualizations are unlikely to

provoke a similar outrage as the examples discussed above, enabling a less heated debate.

In order to prevent erroneous AI-generated figures, we suggest the use of more sophisticated AI-based methods of content creation. Instead of using general-purpose AI tools, researchers and educators should utilize methods that allow them to correct inaccuracies and mistakes. Until more specialized image generators become available, authors should prevent fundamental mistakes by using AI tools that allow the generation of images or 3D models from correct and accurate source materials. While it may be tempting to generate entire figures using simple prompts, the combination of more sophisticated AI tools has the potential to result in more accurate visualizations with little additional effort.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval The article does not contain empirical data.

Informed consent Informed consent does not apply to this theoretical work.

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